

AUTO1 Analytics Engineering Case Study

Detecting Post-Purchase IBAN Changes

Presented by:Uma Obbani

Objective

Build a scalable analytics engineering pipeline to:

- Detect post-purchase bank account changes***
 - Compare old vs new IBAN values***
 - Produce GDPR-safe stakeholder reports***
 - Simulate secure API enrichment architecture***
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Business Problem

Finance UUIDs (bank account references) can change after vehicle purchases. Stakeholders require visibility into when bank account changes occurred, previous and updated IBAN information, and country level impact while maintaining GDPR compliance.

Assumptions & Design Decisions

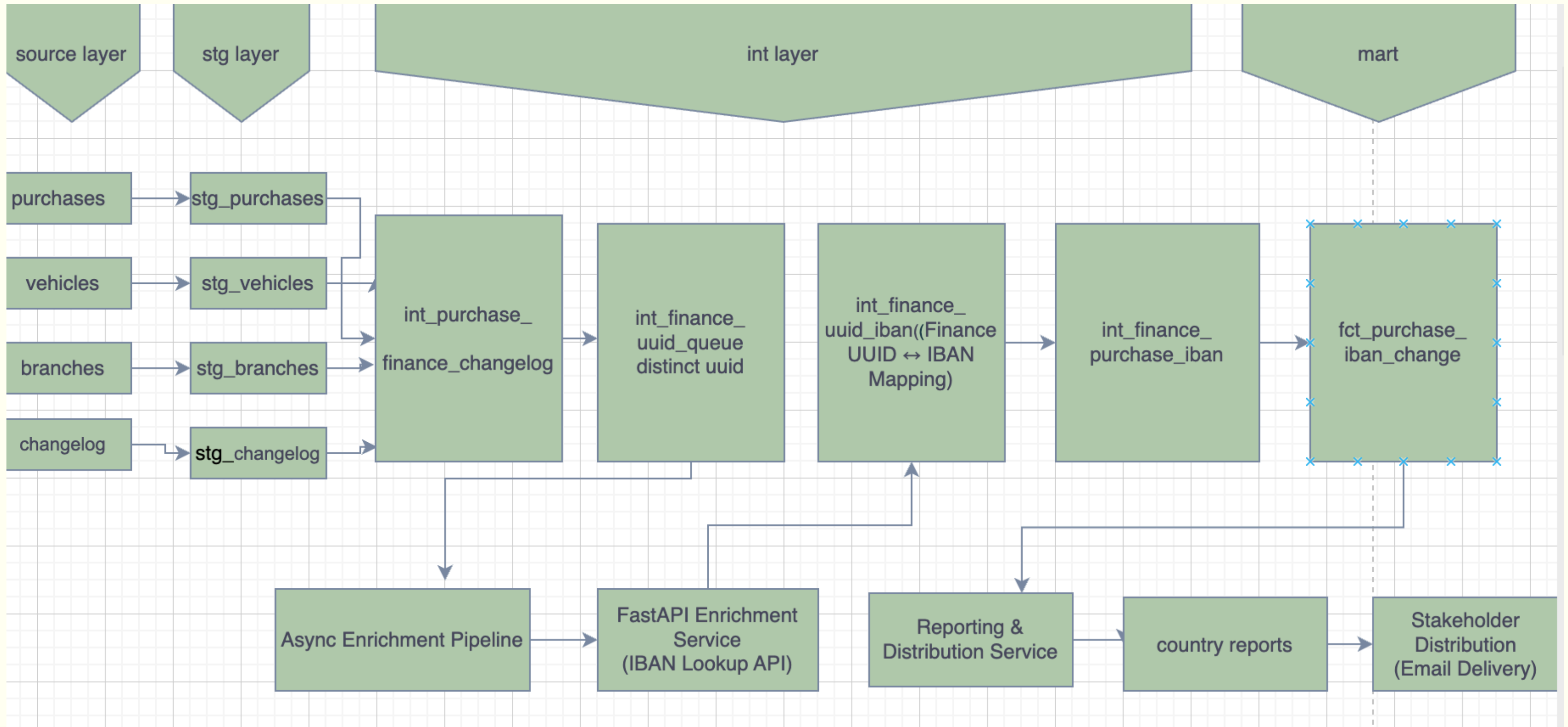
Finance Identifier Selection

- *Purchase data contains the current bank/finance account reference associated with the purchase.*
- *Changelog provides the historical transition between old_finance_uuid → new_finance_uuid.*
- *To identify IBAN changes, both previous and current finance identifiers are required.*
- *Therefore, changelog identifiers were enriched to derive IBAN before and after the change event.*

Country & Branch Selection

- *Country and branch information are prioritized to support accurate stakeholder-level report distribution.*
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Solution Arhitecture



Data Flow & Processing Logic

To support the case study, source datasets were simulated to represent a realistic automotive finance environment

- Purchases – purchase transactions and timestamps***
 - Vehicles – vehicle information associated with purchases***
 - Branches – branch and country information***
 - Changelog – historical Finance UUID change events***
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Data Flow & Processing Logic

Data Standardization

dbt staging models standardize and validate source data before business transformations are applied.

Data Flow & Processing Logic

Business logic:

The intermediate layer consolidates purchase, vehicle, branch, and finance change data into a business ready dataset that identifies valid post purchase finance account changes within the last 12 months.

***Intermediate Layer – Finance UUID Enrichment Queue
Consolidates old and new Finance UUIDs into a deduplicated
enrichment queue.***

Async Enrichment Processing

- *Created a distinct Finance UUID enrichment queue.*
 - *Processed UUIDs asynchronously using Python (asyncio + httpx)*
 - *Simulated an external banking API using FastAPI.*
 - *Retrieved mocked IBAN and banking information.*
 - *Stored enriched results in BigQuery for reporting and change analysis.*
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Fast API

127.0.0.1:8000/docs#/default/search_accounts_search_post

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Axius Technologie...

Aklamio - The Gro...

-H 'x-wa-key: 123abcde123' \

-H 'Content-Type: application/json' \

-d '{

"uuids": [

"a3907caf-51f0-4a95-9e58-756db1a323a2"

]

}

}

Request URL

http://127.0.0.1:8000/search

Server response

Code	Details
200	Response body <pre>{ "results": [{ "uuid": "a3907caf-51f0-4a95-9e58-756db1a323a2", "created_on": "2026-06-01T08:38:39.674560Z", "account": { "holder": "Customer_a3907caf", "iban": "DE28844539536040317115", "bic": "SCFBDE33XXX", "bank_name": "Santander" } }] }</pre> Response headers content-length: 221 content-type: application/ison

IBAN Change Detection Logic

- ***Joined old Finance UUIDs to the enrichment dataset to retrieve the previous IBAN.***
 - ***Joined new Finance UUIDs to the enrichment dataset to retrieve the current IBAN.***
 - ***Compared old and new IBAN values to identify account changes.***
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Mart : Output Dataset

- ***car_id***
- ***branch_id***
- ***country***
- ***purchase_datetime***
- ***old_iban_last4***
- ***old_iban_last_active_datetime***
- ***change_event_datetime***
- ***new_iban_last4***

- ***Reports expose only the last four digits of IBAN values.***
- ***Country-specific stakeholder reports generated automatically.***

clarification :

- ***Report screenshot/reference column mentions bank UUID changed before purchase at***
 - ***Business objective asks for post-purchase bank account / IBAN changes***
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Stake holder reports distribution



maheswaree@gmail.com

to me ▾

Hello,

Please find attached the monthly IBAN Change Reports.

Each Excel file contains IBAN change records for a specific market.

Regards,

Analytics Engineering

20 Attachments • Scanned by Gmail ⓘ ⬇️ 📁 Add all to Drive

	A	B	C	D	E	F	G
1	car_id	branch_id	market	issuetime_before	before_ib_active_dt	after_ib	
2	CAR00054	32	NO	2024-01-1 2804	2024-01-1 8244		
3	CAR00421	32	NO	2024-01-1 3612	2024-01-1 1360		
4	CAR00471	32	NO	2024-01-1 6837	2024-02-1 0541		
5	CAR00491	32	NO	2024-01-1 7031	2024-02-1 5080		
6	CAR00354	32	NO	2024-02-1 3016	2024-04-1 8552		
7	CAR00241	32	NO	2024-02-1 1763	2024-04-1 4119		
8	CAR00031	32	NO	2024-02-1 1262	2024-04-1 0302		
9	CAR00331	32	NO	2024-02-1 3004	2024-04-1 7458		
10	CAR00071	32	NO	2024-03-1 6360	2024-04-1 1531		
11	CAR00031	32	NO	2024-03-1 2529	2024-04-1 5596		
12	CAR00371	32	NO	2024-02-1 4019	2024-04-1 4948		
13	CAR00431	32	NO	2024-03-1 4937	2024-05-1 2252		
14	CAR00071	32	NO	2024-05-1 1549	2024-05-1 8184		
15	CAR00111	32	NO	2024-05-1 1645	2024-05-1 3389		
16	CAR00441	32	NO	2024-05-1 1085	2024-05-1 2655		

	A	B	C	D	E	F	G
1	car_id	branch_id	market	issuetime_before	before_ib_active_dt	after_ib	
2	CAR00541	3	ES	2024-01-1 4810	2024-01-1 8232		
3	CAR00001	3	ES	2024-01-1 4419	2024-01-1 7385		
4	CAR00701	3	ES	2024-01-1 1453	2024-01-1 6460		
5	CAR00601	3	ES	2024-01-1 3048	2024-02-1 1964		
6	CAR00241	3	ES	2024-01-1 2997	2024-02-1 8039		
7	CAR00281	3	ES	2024-01-1 2954	2024-02-1 4481		
8	CAR00341	3	ES	2024-02-1 3617	2024-02-1 0087		
9	CAR00481	3	ES	2024-01-1 7684	2024-03-1 6800		
10	CAR00481	3	ES	2024-01-1 7712	2024-03-1 4509		
11	CAR00441	3	ES	2024-03-1 3138	2024-03-1 1042		
12	CAR00391	3	ES	2024-01-1 5375	2024-04-1 5576		
13	CAR00341	3	ES	2024-01-1 1679	2024-04-1 0628		
14	CAR00321	3	ES	2024-02-1 0348	2024-04-1 0041		
15	CAR00341	3	ES	2024-02-1 9500	2024-04-1 6241		
16	CAR00471	3	ES	2024-04-1 0190	2024-04-1 2894		

	A	B	C	D	E	F	G
1	car_id	branch_id	market	issuetime_before	before_ib_active_dt	after_ib	
2	CAR00204	35	IE	2024-01-1 2096	2024-02-1 6754		
3	CAR00531	35	IE	2024-01-1 4033	2024-02-1 0062		
4	CAR00131	35	IE	2024-01-1 8418	2024-02-1 2378		
5	CAR00131	35	IE	2024-02-1 6795	2024-03-1 3201		
6	CAR00531	35	IE	2024-01-1 0862	2024-03-1 3946		
7	CAR00131	35	IE	2024-01-1 9090	2024-03-1 9531		
8	CAR00401	35	IE	2024-03-1 2381	2024-04-1 1357		
9	CAR00421	35	IE	2024-02-1 6719	2024-04-1 2566		
10	CAR00441	35	IE	2024-02-1 6044	2024-04-1 0352		
11	CAR00564	35	IE	2024-02-1 9030	2024-04-1 3911		
12	CAR00341	35	IE	2024-01-1 6011	2024-04-1 8662		
13	CAR00671	35	IE	2024-04-1 0210	2024-05-1 0985		
14	CAR00421	35	IE	2024-02-1 2644	2024-06-1 5994		
15	CAR00321	35	IE	2024-04-1 5869	2024-06-1 0019		
16	CAR00571	35	IE	2024-05-1 5474	2024-06-1 1618		

Data Quality & Reliability

- *Finance UUID validation*
 - *Duplicate removal before enrichment*
 - *API authentication validation*
 - *Retry failed batches (3 attempts)*
 - *Requested vs returned reconciliation*
 - *dbt uniqueness and null checks*
 - *Branch and country completeness validation*
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Reliability Controls

- *Asynchronous batch processing*
- *Partial failure tracking*
- *Failed batch logging*
- *Full refresh loading strategy (WRITE_TRUNCATE)*

Future Enhancements

- **Incremental processing using
finance_change_timestamp**
 - **IBAN history dimension tracking**
 - **Audit tables**
 - **Monitoring and alerting**
 - **Historical report archival**
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Thank you
